Convergence of the inverse-gamma Newton series

MathDB 357309: proof with an explicit remainder bound

Principia Math In Progress

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Task 1: the full conjecture

Let α be the unique positive zero of the digamma function $\psi = \Gamma'/\Gamma$, let $\beta = \Gamma(\alpha)$, and let $\operatorname{inv}\Gamma_0$ denote the inverse of Γ on $[\alpha, \infty)$. Thus $0 < \beta < 1$. For $n \geq 1$, set

$$c_n = \sum_{k=0}^{n} \frac{k}{\left(\prod_{i=1}^{k} \log \frac{(k+1)!}{i!}\right)\left(\prod_{i=1}^{n-k} \log \frac{(k+1)!}{(k+1+i)!}\right)}. \tag{1}$$

Empty products equal 1.

Theorem 1. *For every real $x > \beta$,*

$$\boxed{\operatorname{inv}\Gamma_0(x) = 2 + \sum_{n=1}^{\infty} c_n \prod_{i=1}^{n} \log \frac{x}{i!}.} \tag{2}$$

The series converges absolutely and locally uniformly on (β, ∞), with the finite inner sums in (1) performed before the outer summation.

More explicitly, put $t = \log x$, $a = \log \beta < 0$, $u_0 = -a$, $b_j = \log((j+1)!)$, and

$$P_N(t) = 2 + \sum_{n=1}^{N} c_n \prod_{j=0}^{n-1} (t - b_j).$$

For every $N \geq 1$,

$$\left|\operatorname{inv}\Gamma_0(e^t) - P_N(t)\right| \leq \frac{|t(t-b_1)|}{b_1(t+u_0)} \prod_{j=2}^{N} \frac{|t-b_j|}{u_0+b_j}. \tag{3}$$

The product in this bound is empty when $N = 1$.

This proves exactly [1, Conjecture 3.1]. Its displayed sum starts at $n = 0$; that term is zero, so (2) has the same coefficients and summation convention.

The analytic input

Define, for real $t > a$,

$$F(t) = \operatorname{inv}\Gamma_0(e^t).$$

We first justify an integral representation of this particular inverse of $\log\Gamma$.

We use the classical conformal mapping theorem for $\log\Gamma$ [2, §2, equation (2)], also proved in [3, Proposition 6.4]. For the branch of $\log\Gamma$ real on the positive axis, the upper half-plane $\mathbb{H}$ is mapped conformally onto

$$\mathcal{V} = \mathbb{C} \setminus \bigcup_{j=0}^{\infty} \big([\log|\Gamma(\alpha_j)|, \infty) - ij\pi\big).$$

Here $\alpha_0 = \alpha$, and for $j \geq 1$ the point $\alpha_j \in (-j, -j+1)$ is the unique zero of ψ in that interval. Every removed ray has nonpositive imaginary part, so $\mathbb{H} \subset \mathcal{V}$. It follows that the inverse of $\log\Gamma$ restricted to $\mathbb{H}$ maps $\mathbb{H}$ into $\mathbb{H}$: it is a Pick function.

For real $t > a$, this inverse has the real boundary value $F(t) > \alpha$. Indeed, $\log\Gamma$ is strictly increasing on (α, ∞) and its derivative there is positive; its local inverse therefore takes nearby points above the real axis to points above the real axis. Uniqueness of the conformal inverse identifies the branches. The local inverse theorem and Schwarz reflection extend F analytically across all of (a, ∞), with real values.

The Herglotz representation and its support characterization [3, p. 90] consequently give

$$F(t) = At + B + \int_{u_0}^{\infty} \left(\frac{u}{1+u^2} - \frac{1}{t+u} \right) d\rho(u), \quad t > a, \tag{4}$$

where $A \geq 0$, $B \in \mathbb{R}$, ρ is a positive measure, and $\int_{u_0}^{\infty} (1+u^2)^{-1}\, d\rho(u) < \infty$. The support follows from real analytic continuation across (a, ∞), followed by the change of variable $u = -s$. Both the conformal mapping theorem and the Herglotz theorem are established inputs, explicitly credited here.

Since $b_0 = 0$, $b_1 = \log 2$, $F(0) = 2$, and $F(b_1) = 3$, subtracting (4) at these two points yields

$$\frac{1}{b_1} = A + \int_{u_0}^{\infty} \frac{d\rho(u)}{u(u+b_1)}. \tag{5}$$

In particular the integral on the right is at most $1/b_1$.

Identifying all coefficients

Let $I_N F$ be the polynomial of degree at most N interpolating F at $b_0, \ldots, b_N$. These are distinct nodes, and

$$F(b_j) = j + 2.$$

The Newton coefficient of order $n \geq 1$ is the divided difference

$$F[b_0, \ldots, b_n] = \sum_{k=0}^{n} \frac{k+2}{\prod_{\substack{0 \leq j \leq n \\ j \neq k}} (b_k - b_j)} = \sum_{k=0}^{n} \frac{k}{\prod_{\substack{0 \leq j \leq n \\ j \neq k}} (b_k - b_j)}.$$

The constant 2 contributes zero because a constant has zero divided differences of positive order; equivalently, compare the leading coefficient in its Lagrange interpolation formula. For $j < k$ and $j > k$, respectively, the denominator factors are

$$b_k - b_j = \log \frac{(k+1)!}{(j+1)!}.$$

Reindexing the first group by $i = j + 1$ and the second by $i = j - k$ gives exactly (1). Thus $I_N F = P_N$, not merely a formal series with matching first coefficients.

The exact remainder

For fixed $u \geq u_0$ let $h_u(t) = -1/(t+u)$. Its interpolation remainder has the rational identity

$$h_u(t) - I_N h_u(t) = \frac{(-1)^N \prod_{j=0}^{N}(t-b_j)}{(t+u)\prod_{j=0}^{N}(u+b_j)}. \tag{6}$$

To verify it, multiply the left side by $t+u$. The result is a polynomial of degree at most $N+1$ vanishing at every b_j and taking the value -1 at $t=-u$. These conditions give precisely the right-hand numerator and sign.

For $N \geq 1$, interpolation reproduces the affine part of (4), as well as the constant term in each regularized integrand. A finite Lagrange sum can be interchanged with the convergent regularized integrals. It follows that

$$F(t) - P_N(t) = (-1)^N \prod_{j=0}^{N}(t-b_j) \int_{u_0}^{\infty} \frac{d\rho(u)}{(t+u)\prod_{j=0}^{N}(u+b_j)}. \tag{7}$$

No separately divergent constant integral has been split off.

For $t > a$ all the denominator factors in (7) are positive. Extract the factors with $j = 0, 1$ and use $u \geq u_0$, $t+u \geq t+u_0 > 0$, and (5). This proves (3):

$$|F(t) - P_N(t)| \leq \frac{|t(t-b_1)|}{t+u_0} \left(\prod_{j=2}^{N} \frac{|t-b_j|}{u_0+b_j} \right) \int_{u_0}^{\infty} \frac{d\rho(u)}{u(u+b_1)}.$$

It also proves the one-sided remainder rule

$$(-1)^N \prod_{j=0}^{N}(t-b_j)\,(F(t) - P_N(t)) \geq 0. \tag{8}$$

Convergence throughout the stated interval

Fix $t > a$. Once $b_j > t$, the product factor in (3) is

$$\frac{|t-b_j|}{u_0+b_j} = 1 - \frac{t+u_0}{u_0+b_j} \quad \text{in } (0,1).$$

Furthermore,

$$\sum_{j=2}^{\infty} \frac{1}{u_0+b_j} = \infty, \tag{9}$$

because $b_j = \log((j+1)!) \leq (j+1)\log(j+1)$ and $\sum_{j\geq 2} 1/(j\log j)$ diverges. The inequality $1 - v \leq e^{-v}$ now shows that the product in (3) tends to zero. If t is a node, interpolation is already exact once that node is included. This proves $P_N(t) \to F(t)$ for every $t > a$, including negative t.

The convergence is uniform on a compact interval $K \subset (a, \infty)$. Choose J such that $b_j > \max K$ for $j \geq J$. The finitely many initial factors and the prefactor in (3) are uniformly bounded on K. In the remaining factors, replace $t+u_0$ by $\min K + u_0 > 0$ and apply (9). This proves local uniform convergence.

For $n \geq 2$, taking divided differences in (4) gives

$$c_n = (-1)^{n-1} \int_{u_0}^{\infty} \frac{d\rho(u)}{\prod_{j=0}^{n}(u+b_j)}. \tag{10}$$

Thus the coefficient signs alternate weakly. For fixed real t, all sufficiently late Newton terms have the same sign, since $t - b_j < 0$ for all sufficiently large j. The already established convergence is therefore absolute. This is also locally uniform absolute convergence: after an index independent of $t \in K$, the absolute tail at each t equals $|F(t) - P_N(t)|$, which tends uniformly to zero. Finally, the substitution $t = \log x$ proves Theorem 1 in full.

Verification and scope

The exact verifier checks the rational-kernel base case, divided-difference induction, remainder update, and regularization cancellation, as well as independent Lagrange constructions at six finite node sets. There are sixteen exact symbolic identities. Independent recursive and printed-product coefficients agree in 24 cases, and 128 computed coefficients have the predicted signs.

Eleven real test points, including four factorial nodes, are checked at degrees $8, 32, 128$ against a separately computed principal inverse and the bound (3), using 220-digit arithmetic with a 160-digit precision comparison. All 33 bound diagnostics pass. These checks do not prove convergence; the all-x argument is above. Convergence can be slow: at $x = 0.89$, the degree-128 error is about 0.182. The theorem asserts convergence, not rapid numerical accuracy.

The complete published source and its latest arXiv version both retain Conjecture 3.1 on page 9. Dated bounded status and literature checks accompany this note. The authoring agent performed the audit; no independent peer review is claimed.

Tasks 2–5: proposed follow-up work

These are four proposed tasks, not additional solved problems.

2. **Task 2.** Determine the leading asymptotic of $F(t) - P_N(t)$ for each fixed nonnode $t > a$, using the behavior of the representing measure near u_0; compare it with the product bound (3).

3. **Task 3.** Determine convergence on the boundary $\Re t = a$ for the corresponding complex Newton series, including its value at the real endpoint $t = a$.

4. **Task 4.** Construct a numerically stable accelerated evaluation of (2), with a certified error tolerance on any prescribed compact subinterval of (β, ∞).

5. **Task 5.** Determine the limit, if it exists, of the Newton interpolation series for $\operatorname{inv}\Gamma_0(x)$ at the unlogged factorial nodes $(j+1)!$, and compare that limit with $\operatorname{inv}\Gamma_0$.

References

[1] D. P. H. Ulgenes, *Series and Product Representations of Gamma, Pseudogamma and Inverse Gamma Functions*, Maple Transactions **5**(1), Article 16090 (2025), 10 pp., doi:10.5206/mt.v5i1.16090. Also arXiv:2301.09699v4.

[2] H. L. Pedersen, *Inverses of gamma functions*, arXiv:1309.2167, 2013, §2, equation (2). Published in Constructive Approximation **41** (2015), 251–267.

[3] H. L. Pedersen, *Pick Functions as a Tool in the Theory of Special Functions*, RIMS Kôkyûroku **1893** (2014), 89–101, complete published survey. See p. 90 for Herglotz representation and support, and pp. 99–100 for the conformal mapping theorem.